



SRM UNIVERSITY

Personalized Quiz Application

(JavaFX-Based Role-Specific Quiz Platform)

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Abstract

The **Personalized Quiz Application** is a role-specific, interactive quiz platform developed using **JavaFX** for the Graphical User Interface (GUI) and **SQLite** as the backend database. The application caters to two primary user groups: **teachers** and **students**, providing distinct functionalities based on user roles.

Teachers can **create quizzes**, manage questions, and monitor student performances, including viewing detailed results. Each quiz is assigned a unique **access code**, ensuring secure and controlled participation. Students, on the other hand, can **register**, **log in**, enter the quiz access code, and attempt quizzes within the stipulated timeframe. Upon submission, students can view their individual scores and track previous attempts in a dedicated **Results** section.

The project emphasizes **user-friendly interface design**, **secure authentication**, and **efficient data management**, aiming to enhance the learning experience by providing a flexible and manageable assessment tool for academic environments.

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1 Introduction

In the digital age, **online assessments** have become a vital component of modern education, providing flexible, scalable, and efficient ways to evaluate student performance. Traditional paper-based examinations are increasingly being replaced by digital platforms that offer **real-time evaluations**, **immediate feedback**, and enhanced **data management**. However, many existing systems lack the flexibility to accommodate different user roles and secure access controls.

The **Personalized Quiz Application** addresses these challenges by providing a **role-specific platform** designed for both **teachers** and **students**. Built using **JavaFX** for a dynamic and responsive **Graphical User Interface (GUI)** and **SQLite** for lightweight yet powerful database management, the application enables teachers to **create quizzes**, manage student participation, and monitor performance, while students can **attempt quizzes**, track their results, and improve their learning outcomes.

This project emphasizes **user authentication**, **secure access via quiz codes**, and a **user-friendly interface**, ensuring that quizzes are conducted in a controlled environment. The design focuses on **modularity**, **scalability**, and **ease of use**, making it adaptable to various educational settings.

The following sections elaborate on the system's **objectives**, **architecture**, **design methodology**, **implementation details**, and **testing**, culminating in a comprehensive overview of the Personalized Quiz Application.

2 Objectives

The main objectives of the **Personalized Quiz Application** are:

- To develop a **role-specific quiz platform** that provides distinct functionalities for **teachers** and **students**.
- To ensure **secure quiz access** through the use of **unique quiz codes**, preventing unauthorized participation.
- To implement a **user-friendly Graphical User Interface (GUI)** using **JavaFX** for an interactive and intuitive user experience.
- To maintain **secure user authentication** mechanisms for **registration** and **login** for both user roles.
- To enable **teachers** to **create quizzes**, manage questions, and monitor **student performance** effectively.

- To allow **students** to **attempt quizzes**, view their scores, and track their progress via a dedicated **Results** section.
- To integrate a **lightweight database system (SQLite)** for efficient data storage and retrieval.
- To ensure the application is **modular**, **scalable**, and adaptable to various **academic environments**.

3 System Requirements

3.1 Hardware Requirements

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Hard Disk:** Minimum 200 MB of free space
- **Display:** Minimum 1024x768 resolution

3.2 Software Requirements

- **Operating System:** Windows 10 or later, Linux, macOS
- **Java Development Kit (JDK):** Version 21 or higher
- **JavaFX SDK:** Version 24.0.1
- **SQLite:** Embedded within the application
- **IDE (Integrated Development Environment):** IntelliJ IDEA / VS Code (recommended for development)
- **Additional Libraries:**
 - SQLite JDBC Driver
 - JavaFX Scene Builder (for GUI design, optional)

4 Technologies Used

The **Personalized Quiz Application** utilizes a range of technologies to ensure efficient development, user-friendly interaction, and reliable data handling:

- **Java:** The primary programming language used for building the application logic and functionalities.
- **JavaFX:** A robust framework for creating responsive and interactive **Graphical User Interfaces (GUI)**.
- **SQLite:** A lightweight, embedded relational database management system used for storing user information, quizzes, questions, and results.
- **SQLite JDBC Driver:** A Java library that enables connectivity between Java applications and the SQLite database.
- **IntelliJ IDEA / Visual Studio Code:** Integrated Development Environments (IDEs) used for coding, debugging, and project management.
- **JavaFX Scene Builder (Optional):** A drag-and-drop GUI design tool for JavaFX applications, aiding in visual interface creation.

These technologies collectively contribute to building a **scalable, maintainable**, and **user-friendly** quiz application, ensuring smooth interaction between the user interface, backend database, and application logic.

5 Workflow

The **Personalized Quiz Application** operates through a structured sequence of processes for both **teachers** and **students**, ensuring seamless interaction between users and the system.

5.1 User Registration and Login

- **Teachers** and **students** can register by providing necessary details like username, password, and role (teacher/student).
- Upon successful registration, users can log in using their credentials.

5.2 Teacher Workflow

- After logging in, teachers access the **Teacher Dashboard**.
- Teachers can **create quizzes** by entering quiz details (title, description).
- For each quiz, teachers can **add questions and multiple-choice options**, marking the correct answer.
- A **unique access code** is automatically generated for each quiz to ensure secure access.
- Teachers can view a list of all created quizzes with details like title, access code, creation date, and status.
- Teachers can also **monitor student performance**, viewing detailed quiz results of all participating students.

5.3 Student Workflow

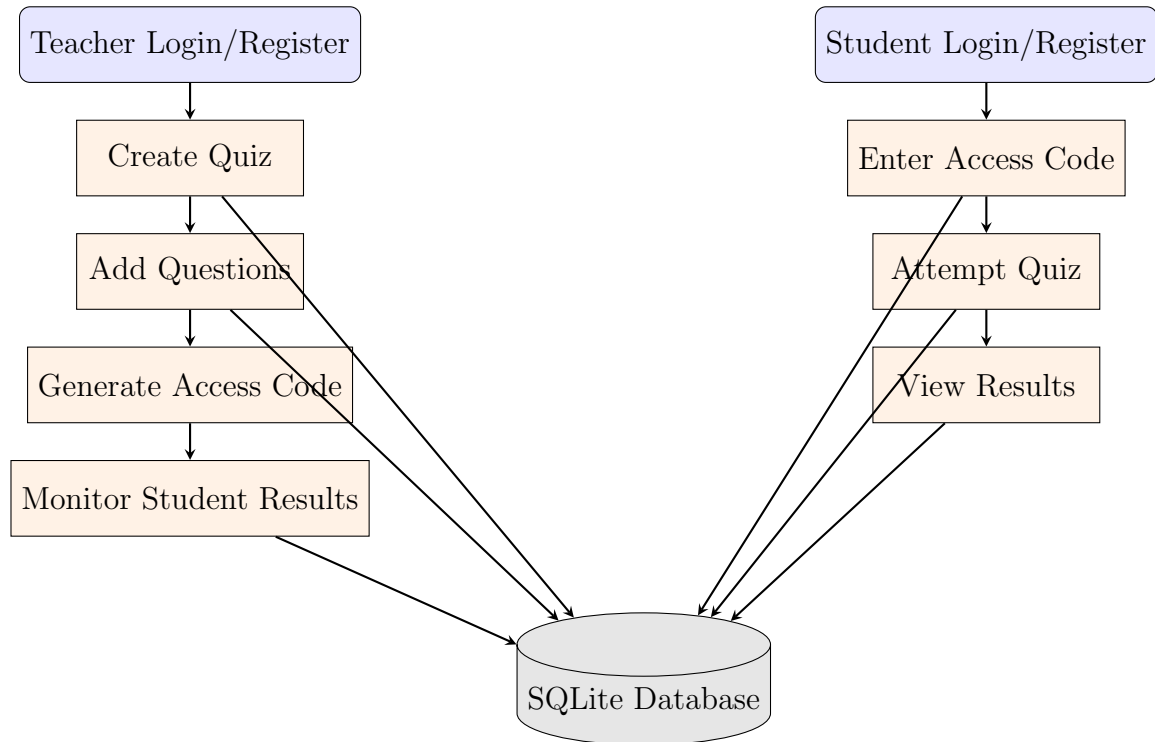
- After logging in, students access the **Student Dashboard**.
- Students enter the **quiz access code** provided by the teacher to join a quiz.
- Once the quiz is available (activated by the teacher), students can **attempt the quiz** by answering multiple-choice questions.
- After submitting, students receive their **quiz score** and can view a list of previously attempted quizzes along with scores and attempt dates in the **Results** section.

5.4 Database Interaction

- The system uses **SQLite** to manage:
 - User credentials and roles.
 - Quiz details and questions.
 - Student results and performance records.

This workflow ensures role-specific functionalities and maintains **secure, controlled access** for quiz participation and result tracking.

5.5 Workflow Diagram



6 Implementation Details

6.1 Key Modules / Classes

- **Login & Registration Module:** Handles user authentication for both **teachers** and **students**. User credentials are stored and verified using **SQLite**.
- **Teacher Dashboard:** Allows teachers to:
 - **Create quizzes** with metadata (title, description).
 - **Add multiple-choice questions** for each quiz.
 - **Generate unique access codes** for secure quiz participation.
 - **Monitor student results** by viewing scores and quiz attempts.
- **Student Dashboard:** Enables students to:
 - **Enter quiz access codes** provided by teachers.

- **Attempt quizzes** by answering multiple-choice questions.
- **View their scores** and **track past attempts**.

6.2 Database Schema

- **Users Table:** Stores user details:
 - user_id, username, password, role
- **Quiz Table:** Stores quiz metadata:
 - quiz_id, quiz_title, access_code, creator_id, created_on
- **Questions Table:** Stores quiz questions:
 - question_id, quiz_id, question_text, option1, option2, option3, option4, correct_option
- **Results Table:** Stores quiz attempts:
 - result_id, student_id, quiz_id, score, attempt_date

6.3 GUI Screens

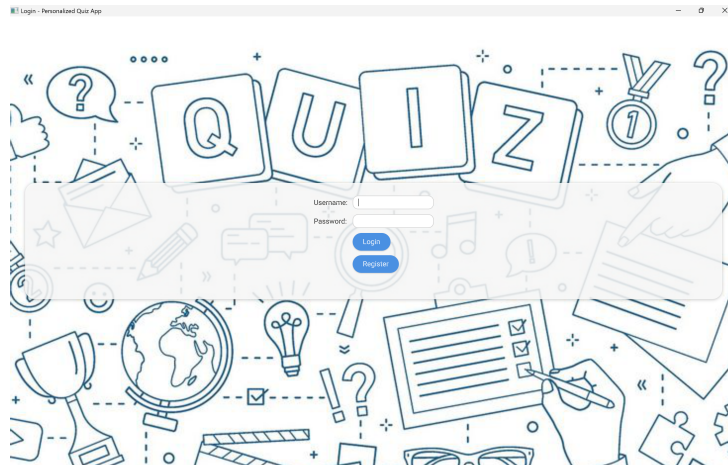
- **Login / Registration Screens**
- **Teacher Panel Screens:**
 - Quiz creation
 - Question addition
 - Results monitoring
- **Student Panel Screens:**
 - Quiz access via code
 - Quiz attempt
 - Results viewing

7 Results

The following screenshots demonstrate the key functionalities and user interfaces of the **Personalized Quiz Application**:

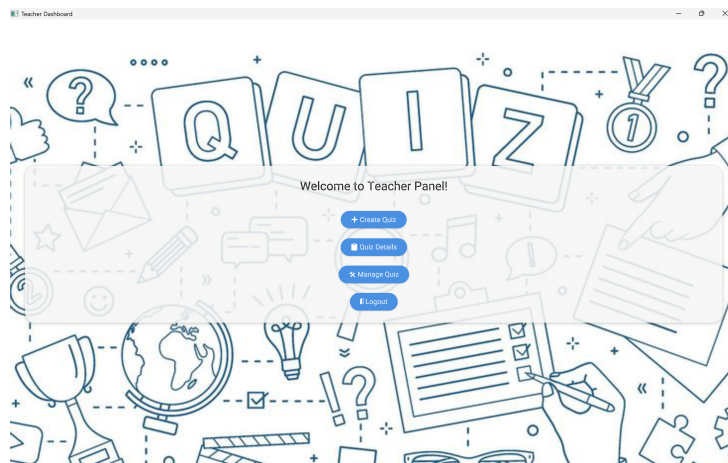
7.1 Login and Registration Interface

This screen allows both **teachers** and **students** to register or log in securely using their credentials.



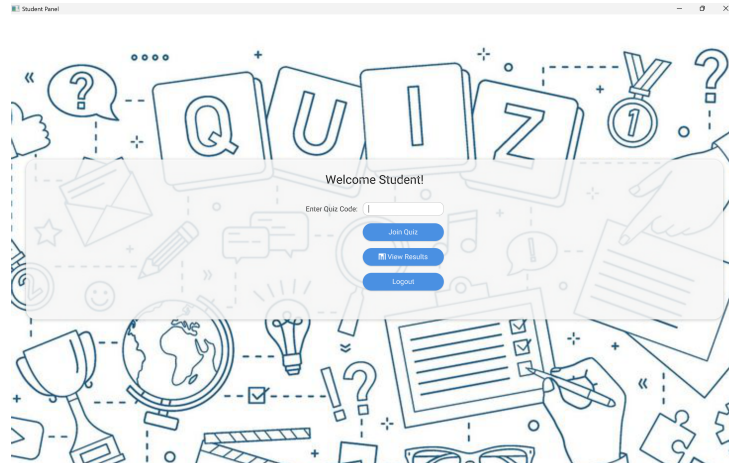
7.2 Teacher Dashboard

After logging in, teachers access the **Teacher Dashboard**, where they can **create quizzes**, **add questions**, and **monitor student results**.



7.3 Student Dashboard

Students access the **Student Dashboard** after logging in. They can **enter quiz access codes**, **attempt quizzes**, and **view their results**.



8 Conclusion

The **Personalized Quiz Application** successfully delivers a **role-specific**, **user-friendly**, and **secure platform** for conducting online assessments. By leveraging **JavaFX** for an interactive **Graphical User Interface (GUI)** and **SQLite** for reliable data management, the application ensures seamless interaction between **teachers** and **students**.

The system empowers **teachers** to create and manage quizzes, monitor student performance, and control quiz access through unique codes. **Students**, in turn, benefit from an intuitive platform to attempt quizzes, view scores, and track their progress over time.

Throughout the development process, the focus remained on **modularity**, **scalability**, and **usability**, making the application adaptable to various **academic environments**. The project demonstrates the effectiveness of combining **JavaFX** and **SQLite** in building practical educational tools that meet modern digital learning needs.

9 Future Scope

While the **Personalized Quiz Application** meets its current objectives effectively, there are several enhancements and extensions that could improve the system's functionality and adaptability:

- **Timed Quizzes:** Implementing a timer for quizzes to enforce time-bound assessments.
- **Quiz Scheduling:** Allowing teachers to schedule quizzes for specific dates and times, improving quiz management and planning.

- **Notifications:** Adding email or in-app notifications for quiz availability, results, and updates.
- **Export Results:** Enabling teachers to export student results in various formats such as PDF or CSV for offline analysis.
- **Mobile Compatibility:** Extending the platform for mobile devices using frameworks like **JavaFXPorts** or migrating to **Android/iOS** platforms for broader accessibility.
- **Enhanced Security:** Implementing encryption mechanisms for sensitive data such as passwords and quiz access codes.
- **Analytical Dashboards:** Integrating analytics to provide insights into student performance trends, quiz difficulty levels, and participation rates.
- **Multi-Language Support:** Extending the platform's usability by offering content in multiple languages.

These potential improvements could further enhance the **usability**, **security**, and **scalability** of the application, making it a more robust tool for educational institutions.

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